Ex:-12:- [B.Q:-10]:- Arun is travelling on his cycle & has calculated to reach point A @ 2pm. If he travels @ 10 kmph. He will reach there @ 12 noon. If he travels @ 15 kmph. At what speed must he travel to reach @ 1pm?

Ans:- t_1 = Travelling @ 10 kmph & reaches @ 2pm

t_2 = " " 15 kmph " " " 12 pm

t_3 = " " ?(x) " " " 1pm.

$$t_1 - t_2 = 2pm - 12pm = \frac{d}{10} - \frac{d}{15} = 2$$

$$\Rightarrow \frac{3d - 2d}{30} = 2 \Rightarrow d = 60km$$

$t_3 - t_2 / t_1 - t_2$ any one we can take to find speed

$$t_3 - t_2 = \frac{d}{x} - \frac{d}{15} = 1pm - 12pm = \frac{60}{x} - \frac{60}{15}$$

$$\Rightarrow 60 - 4x = x \Rightarrow 5x = 60 \Rightarrow \boxed{x = 12 kmph}$$

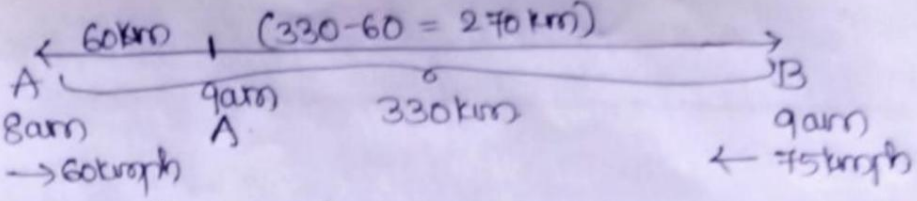
Ex:-13:- [B.Q:-11]:- A & B away from 330km. Annar starts from A @ 8am towards B with 60kmph. Bindu starts from B towards A @ 9am with 75 kmph. At what time will they meet?

Ans:- When two objects moving in opposite direction.

To find out time taken to meet, the objects should start @ same time.

If the objects not starting @ same time towards each other, we have to make them by doing small changes.

(25)



We have to consider their starting time as 9am by excluding A's 1hr journey 60km from 330 and take dist b/w them as 270km.

$$t_{meet} = \frac{270}{R.S(+)} = \frac{270}{60+75} = 2 \text{ hr's}$$

Meeting time = 9am + 2hr's = 11 am

Ex:-14:-[B.Q:-13]:- A train travelling @ 60kmph crosses a another train travelling in the same direction @ 50kmph in 30 seconds. What is combined length of trains?

Ans:- $t_{cross\ each\ other} = 30 = \frac{L_1+L_2}{R.S\ [\text{same direction '-']}}$

$$\Rightarrow 30 = \frac{L_1+L_2}{(60-50)\frac{5}{18}} \Rightarrow L_1+L_2 = 30 \times 10 \times \frac{5}{18_3} = \frac{250}{3} = 83\tfrac{1}{3} \text{ m}$$

Ex:-15:- [B.Q:-15]:- A 300 m long train crosses a platform in 39 seconds, while it crosses a post in 18 seconds. What is the length of the platform?

$$t_{Cross-Platform} = \frac{300+L_p}{S} \rightarrow \text{length of Platform} \quad \therefore = 39. \Rightarrow S = \frac{300+L_p}{39} \rightarrow (2)$$

$$t_{cross-pole} = \frac{300}{S} = 18 \Rightarrow S = \frac{300}{18} = \frac{50}{3} \rightarrow (1)$$

Sub (1) in (2)

$$\Rightarrow \frac{50}{3} = \frac{300+L_p}{39} \Rightarrow \frac{50}{3} \times 39^{13} = 300+L_p$$

$$\Rightarrow L_p + 300 = 650 \Rightarrow L_p = 650 - 300 = 350$$

$$\boxed{L_p = 350 \text{ meters}}$$

Ex:-16:- [B.Q:-16]:- A train of length 270 m with 120 kmph speed crosses another train running in opposite direction @ 80 kmph speed. in 9 seconds. What is 2nd train length?

Ans:- @ $t_{m/cross} = \frac{L_1+L_2}{S_1+S_2} = 9 = \frac{270+L_2}{(120+80)\frac{5}{18}}$

$\Rightarrow 270+L_2 = 9 \times \frac{5}{18} \times 200 \Rightarrow L_2 = 500-270 = \boxed{230 \text{ meters}}$

Ex:-17:-[B.Q:-18]:- The speed of a boat in still water is 22 kmph. & rate of current is 4 kmph. The distance travelled in down-stream in 24 minutes is?

$d_{downstream} = S_{d.s} \times time_{d.s}$

$S_{d.s} = S_b + S_s = 22+4 = 26 \text{ kmph}$

$d_{d.s} = \frac{26}{5.2} \times \frac{24}{60}(hrs) = \boxed{10.4 \text{ km}}$

Ex:-18:-[B.Q:-20]:- A boat running in downstream covers a dist of 22 km in 4 hrs while for covering the same dist upstream it takes 5 hrs. What is speed of the boat in still water (S_b)?

Ans:- $S_w = \frac{S_d - S_u}{2} = \frac{\frac{d_{ds}}{t_{ds}} - \frac{d_{us}}{t_{us}}}{2} = \frac{\frac{22}{4} - \frac{22}{5}}{2}$

$S_w = \frac{5.5-4.4}{2} = \frac{1.1}{2} = \boxed{0.55 \text{ kmph}}$, $S_b = \frac{S_d+S_u}{2}$

$S_b = \frac{5.5+4.4}{2} = \boxed{4.95 \text{ kmph}}$

Ex:-19:- [B.Q:-21]:- A man can row at 5 kmph in still water. If the velocity (speed) of current (water in still water) is 1 kmph. & it takes him 1 hr. to row to a place & back, how far is the place?